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Aim

To write, simulate, and verify Verilog HDL code for the following:

- Basic logic gates using gate-level and dataflow modeling.
- The Boolean expression

$F(A,B,C)=A+BC+ABC$ using gate-level and dataflow modeling.

- Half adder and half subtractor circuits using a test bench in Modelsim.

Tool Required

- Modelsim Software
- Verilog HDL (Hardware Description Language)

Theory

A Hardware Description Language (HDL) is used to describe and model digital systems. Verilog is one such HDL.

- **Modeling Styles:**

- **Gate Level Modeling:** This style describes a circuit by specifying the logic gates (e.g., and, not, or) and how they are interconnected.
- **Dataflow Modeling:** This style focuses on the flow of data between registers and how it is processed. It uses continuous assignments to express logic, often with the

assign keyword.

- **Logic Gates:** These are fundamental electronic circuits that operate on one or more input signals to produce an output signal. Basic gates include AND, OR, and NOT, while NAND and NOR are known as universal gates because any digital circuit can be constructed from them.
- **Half Adder:** This is a combinational circuit designed to add two single-bit numbers. It produces two outputs: a Sum (

$A \oplus B$) and a Carry (AB).

- **Half Subtractor:** This is a digital logic circuit that performs the subtraction of two single-bit binary numbers. It has two outputs: a Difference (

$A \oplus B$) and a Borrow ($A \cdot B$).